
Agent Memory: Vector Search Isn't Enough

- 1 Semantic relevance $\neq$ Contextual intelligence.
- 2 Vector databases struggle with multi-hop reasoning and explicit relationships.
- 3 True agent memory requires structured understanding.

The Bottleneck: Why Vector RAG Fails

- 1 Semantic similarity overgeneralizes, leading to hallucinations on precise queries.
- 2 Query complexity degrades accuracy; 0% for schema-bound queries.
- 3 Lack of explicit relationships makes multi-hop reasoning impossible.

Architectural Fix: Graph RAG

- 1 Knowledge graphs explicitly model entities & relationships.
- 2 Hybrid query processing: vector search for simple, graph traversal for complex.
- 3 Transition-aware memory: independent subtask exploration reduces cost.

Hybrid Query Routing Example

1 Route queries based on complexity: simple vs. multi-hop.

2 Leverage vector search for speed, graph traversal for accuracy.

```
main.py

from langchain import GraphRAG, VectorRAG

graph_rag = GraphRAG(knowledge_graph="legal_graph.kg")
vector_rag = VectorRAG(embedding_model="all-MiniLM-L6-v2")

def hybrid_query(query):
    if is_simple_query(query):
        return vector_rag.search(query)
    else:
        return graph_rag.traverse(query)
```

Performance & Trade-offs

- 1 Graph RAG: 86% accuracy on multi-hop vs. 32% for vector RAG.
- 2 Latency: Vector (ms) vs. Graph (s); hybrid routing mitigates.
- 3 Higher setup cost, but superior accuracy for complex reasoning.

Summary: Build Smarter Agents

- 1 Don't conflate similarity with understanding.
- 2 Implement Graph RAG for robust, contextually intelligent agents.
- 3 Focus on hybrid architectures and transition-aware memory for scale and accuracy.